

OpenShift Virtualization VM Networking

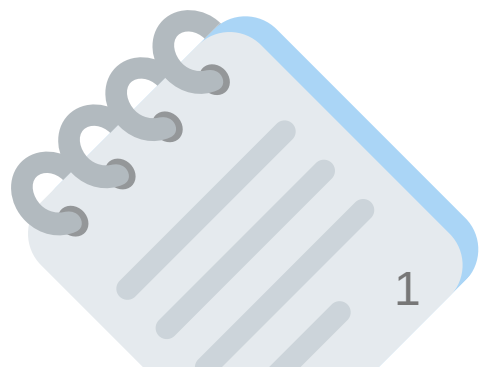
Dale Bewley

Specialist SA

NA West OpenShift 🐱 Team

Red Hat





Container Network Interface

CNI Plugins enable pod network configuration

```
{  
  "cniVersion": "0.3.1",  
  "name": "mynet",  
  "type": 🖱️ <plugin>  
  <parameters>...  
}
```

- 🖱️ **Bridge**
- 🖱️ **ovn-k8s-cni-overlay**
- 🖱️ **SR-IoV**
- 🖱️ **...**



Multus "Meta" CNI Plugin

- The CNI used by OpenShift
- Creates the interfaces in the pod used by the VM

- Is a meta-plugin which executes other plugins

Plugin Chaining

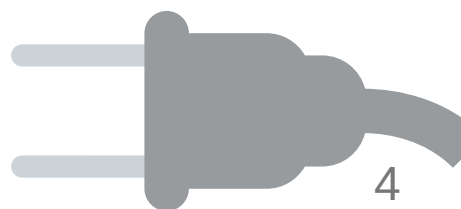


CNI Plugins

There are many plugins and they may be chained by Multus

```
$ oc rsh -n openshift-multus \  
  multus-2tt2c ls -1 /var/lib/cni/bin  
bandwidth  
bond  
bridge  
cert-approver  
cnv-bridge  
cnv-tuning  
dhcp  
dummy  
egress-router  
firewall
```

```
host-device  
host-local  
install_multus  
ip-control-loop  
ipvlan  
kubeconfig_generator  
loopback  
macvlan  
multus  
multus-daemon  
multus-shim  
network-passt-binding  
ovn-k8s-cni-overlay  
...
```



bridge Plugin

Bridge can be used to attach to a Linux Bridge which may enable VLAN access or [VGT](#)

```
{  
  "cniVersion": "0.3.1",  
  "name": "mynet",  
  "type": "bridge",  
  ...  
}
```

- 💡 Don't use cnv-bridge, use bridge. They are identical.



ovn-k8s-cni-overlay Plugin

```
{  
  "cniVersion": "0.3.1",  
  "name": "mynet",  
  "type": "ovn-k8s-cni-overlay",  
  "topology":  <topology>,  
  ...  
}
```



Primary Versus Secondary Networks

Pod Primary Network Interface

- Default route
- CNI cluster default
- IPAM:
 - "infrastructure locked" Cluster Network (10.128.0.0/14)
 - Or a Primary User Defined Network for the NS
- Traffic filtering: `NetworkPolicy`

Pod Secondary Network Interface

- Types:
- IPAM: (depends)
 - "infrastructure locked" Cluster Network (10.128.0.0/14)
 - Or a Primary User Defined Network for the NS
- Traffic filtering: `MultiNetworkPolicy`
-



KubeVirt Interfaces and Networks

- ❶ Primary Cluster Network

- ❷ Primary User Defined Networks

- ❸ Secondary Localnet (VLANs)

- ❹ Secondary User Defined Networks



Network attachments available for Virtual Machines on OpenShift with example implementations



VM Examples - Cluster Network

All VMs have the same address internally and masquerade using the pod IP.

```
# VM attached to default cluster network
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: vm-on-cluster-network
spec:
  template:
    spec:
      domain:
        devices:
          interfaces:
            - macAddress: '02:86:5e:00:00:07'
              masquerade: {}
              model: virtio
              name: default
      networks:
        - name: default
          pod: {}
```

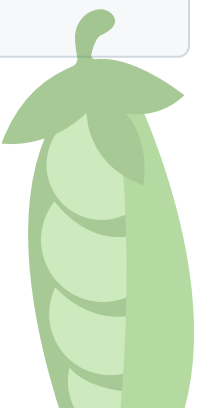
Virt-Launcher Pod

Two ethernet interfaces in the virt launcher pod.

- Infrastructure locked `10.128.0.0/14` cluster network
- Always on `10.0.2.1/24`
- `k6t-eth0` short for `kubevirt-eth0` is a bridge enslaving `tap0`
- `tap0` is passed to QEMU for `eth0` in the VM

```
sh-5.1$ ip -c link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: eth0@if379: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1400 qdisc noqueue state UP mode DEFAULT group default
    link/ether 0a:58:0a:83:01:61 brd ff:ff:ff:ff:ff:ff link-netnsid 0
3: k6t-eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1400 qdisc noqueue state UP mode DEFAULT group default qlen 1000
    link/ether 02:00:00:00:00:00 brd ff:ff:ff:ff:ff:ff
4: tap0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1400 qdisc fq_codel master k6t-eth0 state UP mode DEFAULT group default qlen 1000
    link/ether be:53:ae:c8:c5:66 brd ff:ff:ff:ff:ff:ff

sh-5.1$ ip -br -c -4 a
lo                UNKNOWN          127.0.0.1/8
eth0@if379        UP                10.131.1.97/23
k6t-eth0          UP                10.0.2.1/24
```



Virtual Machine

- VM *always* has IP `10.0.2.2/24` .
- Masquerades as pod IP `10.131.1.97/23` above.

```
[cloud-user@vm-pod ~]$ ip -c link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1400 qdisc fq_codel state UP mode DEFAULT group default qlen 1000
    link/ether 02:86:5e:00:00:10 brd ff:ff:ff:ff:ff:ff
    altname enp1s0

[cloud-user@vm-pod ~]$ ip -br -c -4 a
lo                UNKNOWN          127.0.0.1/8
eth0              UP                10.0.2.2/24

[cloud-user@vm-pod ~]$ ip -c route
default via 10.0.2.1 dev eth0 proto dhcp src 10.0.2.2 metric 100
10.0.2.0/24 dev eth0 proto kernel scope link src 10.0.2.2 metric 100
```

VM on Default Cluster Network

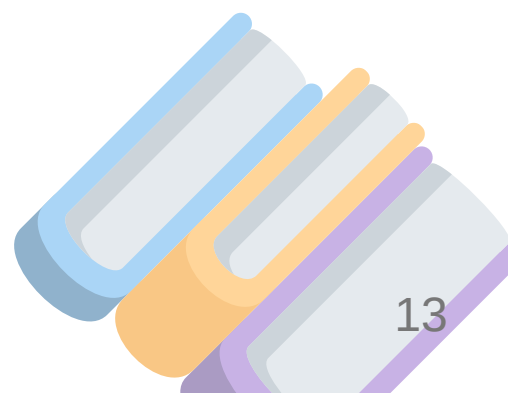
Summary

Virt-launcher Pods

- eth0 on cluster network `10.128.0.0/14`
- k6t-eth0 always has IP `10.0.2.1/24`

VirtualMachines

- eth0 is always IP `10.0.2.2/24`
- Default gateway is always `10.0.2.1` on virt-launcher pod
- Masquerades at pod edge as IP of the virt-launcher pod
- Masquerades at node edge as IP of node default interface `br-ex`



VM Examples - Primary UDN

VMs have unique IPs from UDN subnet

Only Layer2 topology is supported (*localnet* in 4.19)

```
apiVersion: k8s.ovn.org/v1
kind: UserDefinedNetwork
  name: primary-udn
spec:
  topology: Layer2
  layer2:
    ipam:
      lifecycle: Persistent
      role: Primary
      subnets:
        - 10.1.1.0/24
```

```
# VM attached to primary UDN
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: vm-on-primary-udn
spec:
  template:
    spec:
      domain:
        devices:
          interfaces:
            - binding:
                name: l2bridge
              model: virtio
              name: default
      networks:
        - name: default
          pod: {}
```

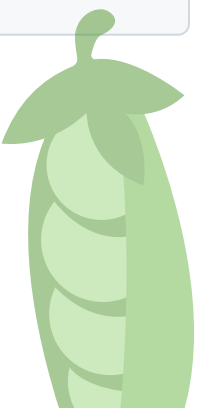
Virt-Launcher Pod

Two ethernet interfaces in the virt launcher pod.

- Infrastructure locked `10.128.0.0/14` cluster network for kubelet health checks *only*
- Unique IP on the UDN range `10.1.1.0/24`

```
sh-5.1$ ip -c link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: eth0@if356: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1400 qdisc noqueue state UP mode DEFAULT group default
    link/ether 0a:58:0a:83:01:4b brd ff:ff:ff:ff:ff:ff link-netnsid 0
3: ovn-udn1-nic@if357: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1400 qdisc noqueue master k6t-ovn-udn1 state UP mode DEFAULT group default
    link/ether 06:1b:c3:df:4d:d3 brd ff:ff:ff:ff:ff:ff link-netnsid 0
4: k6t-ovn-udn1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1400 qdisc noqueue state UP mode DEFAULT group default qlen 1000
    link/ether 06:1b:c3:df:4d:d3 brd ff:ff:ff:ff:ff:ff
5: tap0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1400 qdisc fq_codel master k6t-ovn-udn1 state UP mode DEFAULT group default qlen 1000
    link/ether 82:38:45:e8:1a:3d brd ff:ff:ff:ff:ff:ff
6: ovn-udn1: <BROADCAST,NOARP> mtu 1400 qdisc noop state DOWN mode DEFAULT group default qlen 1000
    link/ether 0a:58:0a:01:01:03 brd ff:ff:ff:ff:ff:ff

sh-5.1$ ip -br -c -4 a
lo                UNKNOWN          127.0.0.1/8
eth0@if356        UP                10.131.1.75/23
ovn-udn1          DOWN             10.1.1.3/24
```



Virtual Machine

- One ethernet interface in the VM with IP from primary UDN

```
[cloud-user@vm-primary-udn ~]$ ip -c link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1400 qdisc fq_codel state UP mode DEFAULT group default qlen 1000
    link/ether 0a:58:0a:01:01:03 brd ff:ff:ff:ff:ff:ff
    altname enp1s0

[cloud-user@vm-primary-udn ~]$ ip -br -c -4 a
lo                UNKNOWN          127.0.0.1/8
eth0              UP                10.1.1.3/24

[cloud-user@vm-primary-udn ~]$ ip -c route
default via 10.1.1.1 dev eth0 proto dhcp src 10.1.1.3 metric 100
10.1.1.0/24 dev eth0 proto kernel scope link src 10.1.1.3 metric 100
```

VM on Primary User Defined Network

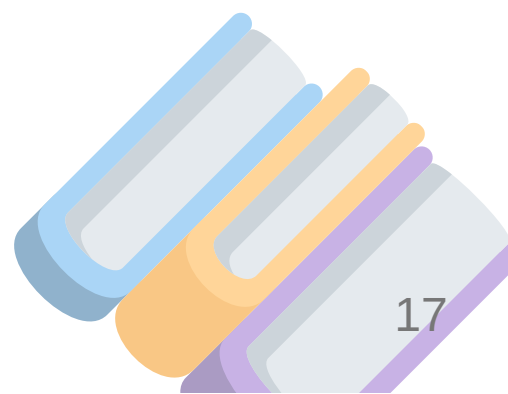
Summary

Virt-Launcher Pod

- Two ethernet interfaces
- eth0@if356 is on infrastructure locked cluster network `10.128.0.0/14`
- ovn-udn1 is on primary UDN `10.1.1.3/24`

Virtual Machine

- eth0 has unique IP `10.1.1.3/24` from primary UDN of this Namespace
- Default gateway is `10.1.1.1`
- Masquerades at UDN gateway router as the IP from `169.254.0.0/17` associated with the UDN
- Masquerades at node edge as IP of node default interface `br-ex`



Masquerade Subnet

Each UDN has two IPs allocated from the masquerade subnet

`169.254.0.0/17` .

```
# oc get network.operator/cluster -o yaml
apiVersion: operator.openshift.io/v1
kind: Network
metadata:
  name: cluster
spec:
  clusterNetwork:
  - cidr: 10.128.0.0/14
    hostPrefix: 23
  defaultNetwork:
    ovnKubernetesConfig:
      egressIPConfig: {}
      gatewayConfig:
        ipv4: {} # <-- default: 169.254.0.0/17
        ipv6: {} # <-- default: fd69::/112
        routingViaHost: false
```



VM Examples - Primary and Secondary UDN

```
apiVersion: k8s.ovn.org/v1
kind: UserDefinedNetwork
  name: primary-udn
spec:
  topology: Layer2
  layer2:
    ipam:
      lifecycle: Persistent
    role: Primary
    subnets:
      - 10.1.1.0/24
```

```
apiVersion: k8s.ovn.org/v1
kind: UserDefinedNetwork
  name: secondary-udn
spec:
  topology: Layer2
  layer2:
    ipam:
      lifecycle: Persistent
    role: Secondary
    subnets:
      - 10.2.2.0/24
```

```
# VM attached to primary UDN and secondary UDN
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: vm-on-primary-udn
spec:
  template:
    spec:
      domain:
        devices:
          interfaces:
            - binding:
                name: l2bridge
                model: virtio
                name: default
            - bridge: {}
              macAddress: '02:86:5e:00:00:0a'
              model: virtio
              name: secondary-udn
      networks:
        - name: default
          pod: {}
        - multus:
            networkName: secondary-udn
            name: secondary-udn
```



Virt-Launcher Pod

```
sh-5.1$ ip -c link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: eth0@if412: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1400 qdisc noqueue state UP mode DEFAULT group default
   link/ether 0a:58:0a:83:01:80 brd ff:ff:ff:ff:ff:ff link-netnsid 0
3: ovn-udn1-nic@if413: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1400 qdisc noqueue master k6t-ovn-udn1 state UP mode DEFAULT group default
   link/ether 92:16:66:87:e3:d3 brd ff:ff:ff:ff:ff:ff link-netnsid 0
4: 2eae7330186-nic@if414: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1400 qdisc noqueue master k6t-2eae7330186 state UP mode DEFAULT group default
   link/ether 26:fc:0d:92:fe:71 brd ff:ff:ff:ff:ff:ff link-netnsid 0
5: k6t-ovn-udn1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1400 qdisc noqueue state UP mode DEFAULT group default qlen 1000
   link/ether 92:16:66:87:e3:d3 brd ff:ff:ff:ff:ff:ff
6: tap0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1400 qdisc fq_codel master k6t-ovn-udn1 state UP mode DEFAULT group default qlen 1000
   link/ether ea:a8:63:f5:7c:f7 brd ff:ff:ff:ff:ff:ff
7: ovn-udn1: <BROADCAST,NOARP> mtu 1400 qdisc noop state DOWN mode DEFAULT group default qlen 1000
   link/ether 0a:58:0a:01:01:03 brd ff:ff:ff:ff:ff:ff
8: k6t-2eae7330186: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1400 qdisc noqueue state UP mode DEFAULT group default qlen 1000
   link/ether 26:fc:0d:92:fe:71 brd ff:ff:ff:ff:ff:ff
9: tap2eae7330186: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1400 qdisc fq_codel master k6t-2eae7330186 state UP mode DEFAULT group default qlen 1000
   link/ether 32:64:42:71:34:80 brd ff:ff:ff:ff:ff:ff
10: pod2eae7330186: <BROADCAST,NOARP> mtu 1400 qdisc noop state DOWN mode DEFAULT group default qlen 1000
    link/ether 02:00:0a:02:02:03 brd ff:ff:ff:ff:ff:ff

sh-5.1$ ip -br -c -4 a
lo                UNKNOWN          127.0.0.1/8
eth0@if412        UP                10.131.1.128/23
ovn-udn1          DOWN             10.1.1.3/24
k6t-2eae7330186   UP                169.254.75.11/32
pod2eae7330186    DOWN             10.2.2.1/24
```



Virtual Machine

```
[cloud-user@vm-secondary-udn ~]$ ip -c link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1400 qdisc fq_codel state UP mode DEFAULT group default qlen 1000
    link/ether 0a:58:0a:01:01:03 brd ff:ff:ff:ff:ff:ff
    altname enp1s0
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1400 qdisc fq_codel state UP mode DEFAULT group default qlen 1000
    link/ether 02:00:0a:02:02:03 brd ff:ff:ff:ff:ff:ff
    altname enp2s0

[cloud-user@vm-secondary-udn ~]$ ip -br -c -4 a
lo                UNKNOWN          127.0.0.1/8
eth0              UP                10.1.1.3/24
eth1              UP                10.2.2.1/24

[cloud-user@vm-secondary-udn ~]$ ip -c route
default via 10.1.1.1 dev eth0 proto dhcp src 10.1.1.3 metric 100
10.1.1.0/24 dev eth0 proto kernel scope link src 10.1.1.3 metric 100
10.2.2.0/24 dev eth1 proto kernel scope link src 10.2.2.1 metric 101
```

VM on Primary & Secondary User Defined Network

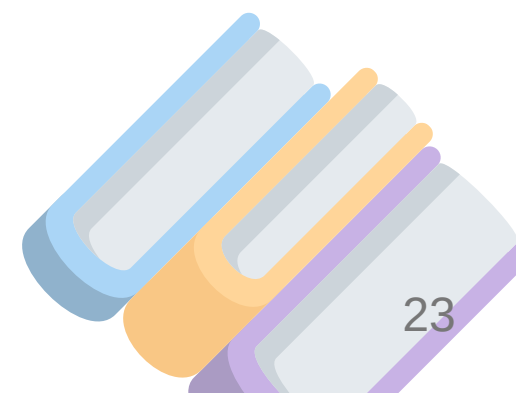
Summary

Virt-Launcher Pod

- Three ethernet interfaces
- eth0@if412 is on infrastructure locked cluster network `10.128.0.0/14`
- ovn-udn1 is on primary UDN `10.1.1.3/24`
- pod2eae7330186 is on secondary UDN `10.2.2.1/24`

Virtual Machine

- eth0 has unique IP `10.1.1.3/24` from primary UDN of this Namespace
- eth1 has unique IP `10.2.2.1/24` from secondary UDN of this Namespace
- Default gateway is `10.1.1.1`
- Masquerades at UDN gateway router as the IP from `169.254.0.0/17` for the UDN
- Masquerades at node edge as IP of node default interface `br-ex`



VM Examples - Localnet Secondary

VMs directly attached to VLAN

No UDN support for localnet topology as of 4.18.9.

Net-attach-def only.

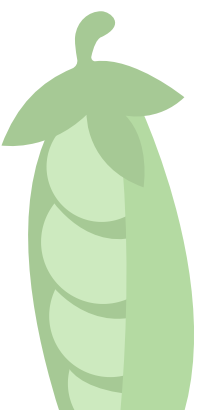
```
apiVersion: k8s.cni.cncf.io/v1
kind: NetworkAttachmentDefinition
metadata:
  name: demo-vlan-1924
spec:
  config: |-
    {
      "cniVersion": "0.4.0",
      "name": "demo-vlan-1924",
      "type": "ovn-k8s-cni-overlay",
      "topology": "localnet",
      "netAttachDefName": "demo-vm-localnet/demo-vlan-1924",
      "vlanID": 1924,
      "ipam": {}
    }
```

```
# VM attached to secondary localnet
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: vm-localnet
spec:
  template:
    spec:
      domain:
        devices:
          interfaces:
            - macAddress: '02:86:5e:00:00:13'
              masquerade: {}
              model: virtio
              name: default
            - bridge: {}
              macAddress: '02:86:5e:00:00:14'
              model: virtio
              name: nic-vlan-1924
      networks:
        - name: default
          pod: {}
        - multus:
            networkName: demo-vlan-1924
            name: nic-vlan-1924
```

Virt-Launcher Pod

```
sh-5.1$ ip -c link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: eth0@if607: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1400 qdisc noqueue state UP mode DEFAULT group default
    link/ether 0a:58:0a:83:00:54 brd ff:ff:ff:ff:ff:ff link-netnsid 0
3: 16711a0a730-nic@if608: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1400 qdisc noqueue master k6t-16711a0a730 state UP mode DEFAULT group default
    link/ether 12:6f:ef:f4:36:9a brd ff:ff:ff:ff:ff:ff link-netnsid 0
4: k6t-eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1400 qdisc noqueue state UP mode DEFAULT group default qlen 1000
    link/ether 02:00:00:00:00:00 brd ff:ff:ff:ff:ff:ff
5: tap0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1400 qdisc fq_codel master k6t-eth0 state UP mode DEFAULT group default qlen 1000
    link/ether 8a:47:bc:d4:8b:d0 brd ff:ff:ff:ff:ff:ff
6: k6t-16711a0a730: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1400 qdisc noqueue state UP mode DEFAULT group default qlen 1000
    link/ether 12:6f:ef:f4:36:9a brd ff:ff:ff:ff:ff:ff
7: tap16711a0a730: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1400 qdisc fq_codel master k6t-16711a0a730 state UP mode DEFAULT group default qlen 1000
    link/ether 7a:f1:ab:eb:e9:e1 brd ff:ff:ff:ff:ff:ff
8: pod16711a0a730: <BROADCAST,NOARP> mtu 1400 qdisc noop state DOWN mode DEFAULT group default qlen 1000
    link/ether 02:86:5e:00:00:16 brd ff:ff:ff:ff:ff:ff

sh-5.1$ ip -br -c -4 a
lo                UNKNOWN          127.0.0.1/8
eth0@if607        UP                10.131.0.84/23
k6t-eth0          UP                10.0.2.1/24
```



Virtual Machine

```
[cloud-user@vm-localnet ~]$ ip -c link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1400 qdisc fq_codel state UP mode DEFAULT group default qlen 1000
    link/ether 02:86:5e:00:00:15 brd ff:ff:ff:ff:ff:ff
    altname enp1s0
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1400 qdisc fq_codel state UP mode DEFAULT group default qlen 1000
    link/ether 02:86:5e:00:00:16 brd ff:ff:ff:ff:ff:ff
    altname enp2s0
```

```
[cloud-user@vm-localnet ~]$ ip -br -c -4 a
lo                UNKNOWN          127.0.0.1/8
eth0              UP                10.0.2.2/24
eth1              UP                192.168.4.71/24
```

```
[cloud-user@vm-localnet ~]$ ip -c route
default via 10.0.2.1 dev eth0 proto dhcp src 10.0.2.2 metric 100
default via 192.168.4.1 dev eth1 proto dhcp src 192.168.4.71 metric 101
10.0.2.0/24 dev eth0 proto kernel scope link src 10.0.2.2 metric 100
192.168.4.0/24 dev eth1 proto kernel scope link src 192.168.4.71 metric 101
```

VM on Localnet

Summary

Virt-launcher Pod

- eth0@if607 is on the cluster network `10.128.0.0/14`
- tap0 is passed to QEMU for eth0
- tap16711a0a730 is passed to QEMU for eth1

Virtual Machine

- eth0 is always IP `10.0.2.2/24`
- Default gateway is always `10.0.2.1` on virt-launcher
- Masquerades at node edge as IP of node default interface `br-ex`
- eth1 is `192.168.4.71/24` from DHCP on datacenter VLAN 1924



VM Examples - Localnet Secondary UDN

VMs directly attached to VLAN

TBD

 Support for UDN localnet topology is targeted for 4.19.

Q&A / further reading

- [CNI Spec](#)

References

CNI spec & OVN-Kubernetes overview

2. OVN-Kubernetes (Primary CNI)

- **ovn-kubernetes GitHub**
 - Website: ovn-kubernetes.io/
 - Repo: [ovn-org/ovn-kubernetes](https://github.com/ovn-org/ovn-kubernetes)
 - Core repo for the OVN-Kubernetes integration: CNI binaries, controllers, docs.
- **OpenShift OVN-Kubernetes Guide**
 - Docs: [OpenShift Container Platform Networking – OVN-Kubernetes](#)
 - Red Hat's overview of